

Variable Selection Part 2

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How many covariates can I include?

It is fairly common in Biomedical data that we have many covariates, and we need to ask whether our model can hold its predictive value given the number of cases that we have at hand. Thus we need to know how to calculate a suitable sample size (n) relative to the number of covariates p for a reliable prediction.

In this lecture we will revise the multivariate prediction models for continuous outcomes.

Ad hoc solution

The following guidelines are presented by Harrell (2001).

In the text, it is claimed that according to several studies validated on independent data sets and it was concluded that a fitted regression model is likely to be reliable when the number of predictors p is less than $m/10$ or $m/20$ where m is the **limiting sample size**.

Type of Response Variable	Limiting sample size m
Continuous	n (total sample size)
Binary	$\min(n_1, n_2)$ where n_i represents the size of each category.
Ordinal (k categories)	$n - \frac{1}{n^2} \sum_{i=1}^k n_i^3$

Note that p represents all the variables screened for association with the response, including nonlinear terms and interactions.

Example.

Let's suppose that we have a continuous output variable and our number of covariates is $p = 30$, and all of them are continuous. We modeled our data with no interactions or quadratic terms. In this case, the limiting sample size is the total number of *rows* in our data set, thus we need at least 300 to 600 to possibly avoid overfitting. If we include interactions that number can expand quickly, say if we consider all interactions $\binom{p}{2} = 435$, plus the number of original variables $p = 30$, the number of rows that we need are at least 4650 if we are conservative in our estimates.

More recent estimates

We need to revise some concepts.

Calibration plot represents the observed response against predicted responses. Thus, ideally when you have a very good model the slope is closer to 1. The *slope* of the calibration curve is called the **shrinkage factor** (S).

A heuristic shrinkage estimate

$$\hat{S} = \frac{\text{model } \chi^2 - p}{\text{model } \chi^2}$$

where p is the number of degrees of freedom.

Multilinear regression setup

We will follow the Riley et al. (2019) paper. First, we consider the model

$$E(Y_i) = \alpha + \sum_i \hat{\beta}_i X_i$$

and to adjust for overfitting, we use

$$E(Y_i) = \alpha^* + S \left(\sum_i \hat{\beta}_i X_i \right)$$

where α^* is the revised intercept (re-estimated to ensure that the overall predicted mean agrees with the observed mean) and we include the estimate for the shrinkage S .

There is an alternative expression to shrinkage, the so-called Copas closed-form S_C

$$S_C = 1 + \frac{p-2}{n \ln \left(1 - \left(\frac{R_{adj}^2(n-p-1)+p}{n-1} \right) \right)}$$

This formula has some advantages, since we can establish setup values for the shrinkage and the R_{adj}^2 , so we can obtain n as a function of p .

A shrinkage of 0.9 is considered to be the minimum necessary to avoid overfitting.

Variance explained by the new model

In the paper, there are more technical conditions for the selection of the number of predictors.

The estimation of the R_{app}^2 which is the **apparent** estimate of the prediction model R^2 in the dataset used to develop the model. A second criteria that considers that the difference δ between R_{app}^2 and R_{adj}^2 is small to account for the optimism in the developed model's apparent proportion of variance explained is small.

$$\delta = \frac{p(1 - R_{adj}^2)}{n - 1}$$

Thus

$$n \geq 1 + \frac{p(1 - R_{adj}^2)}{\delta}$$

This may be known from the literature R_{app}^2 and if we hope to have an R_{adj}^2 not far off more than 10%, we can estimate the number of cases needed.

More desiderata

Riley et al. has a couple of criteria that are important from the statistical perspective

- Precise estimate of the residual standard deviation.
- Precise estimate of the model intercept.

References

- Harrell, F. E. Jr. 2001. *Regression Modeling Strategies: With Applications to Linear Models, Logistic Regression and Survival Analysis*. 1st. edition. Springer Series in Statistics. Springer.
- Riley, Richard D., Kym I. E. Snell, Joie Ensor, Danielle L. Burke, Frank E. Harrell Jr, Karel G. M. Moons, and Gary S. Collins. 2019. “Minimum Sample Size for Developing a Multi-variable Prediction Model: Part i – Continuous Outcomes.” *Statistics in Medicine* 38 (7): 1262–75. <https://doi.org/https://doi.org/10.1002/sim.7993>.